

**ANKARA YILDIRIM BEYAZIT UNIVERSITY
GRADUATE SCHOOL OF NATURAL AND APPLIED
SCIENCES**



THESIS ss

**M.Sc. Thesis by
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September, 2023

ANKARA

THESIS ss

**A Thesis Submitted to the
Graduate School of Natural And Applied Sciences of
Ankara Yildirim Beyazit University
In Partial Fulfillment of the Requirements for the Degree of Master of Science in
Computer Engineering, Department of Computer Engineering**

**by
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September, 2023

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M.Sc. THESIS EXAMINATION RESULT FORM

We have read the thesis entitled "**THESIS ss**" completed by **MUHAMMED YASINHAN YAŞAR** under supervision of **Assoc. Prof. MUHAMMED ABDULLAH BÜLBÜL** and we certify that in our opinion it is fully adequate, in scope and in quality, as a thesis for the degree of Master of Science.

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ACKNOWLEDGEMENTS

Firstly, I would like to express my sincere gratitude to my advisor Prof. Asds ASDAS for continuous support of...

Beside my advisor, I am deeply indebted to Prof. Asdas

I thank my colleagues at AYBU...

September, 2023

Muhammed Yasinhan YAŞAR

THESIS ss**ABSTRACT**

Imaging in sub-Terahertz (THz) band has become an attractive research area with emerging technology since it has a wide range of potential applications in defense, security or airborne systems...

Keywords: Compressed sensing, sub-THz, single pixel imaging, super-resolution, computational ghost imaging, millimeter wave, image quality assessment, convex optimization, matching pursuit, hadamard.

TÜRKÇE TEZ BAŞLIĞITÜRKÇE TEZ BAŞLIĞITÜRKÇE TEZ BAŞLIĞITÜRKÇE TEZ BAŞLIĞITÜRKÇE TEZ BAŞLIĞI ÖZ

Terahertz (THz)-altı bandında görüntüleme, savunma, güvenlik veya havacılık sistemlerinde çok çeşitli potansiyel uygulamalara sahip olması nedeniyle, gelişen teknolojiyle birlikte çekici bir araştırma alanı haline gelmiştir...

Anahtar kelimeler: Sıkıştırılmış algılama, THz-altı, tek piksel görüntüleme, süper-çözünürlük, hesaplamalı hayalet görüntüleme, milimetre dalga, görüntü kalitesi değerlendirme, konveks optimizasyon, eşleştirme algoritması, hadamard.

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NOMENCLATURE

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CHAPTER 1

INTRODUCTION

Bu döküman Ankara Yıldırım Beyzıt Üniversitesi Fen Bilimleri Enstitüsünün Haziran 2021 kurallarına göre güncellenmiştir.

1.1 Section Title

This is a text inside the section.

1.1.1 Subsection title

After subsection text.

Example Figure

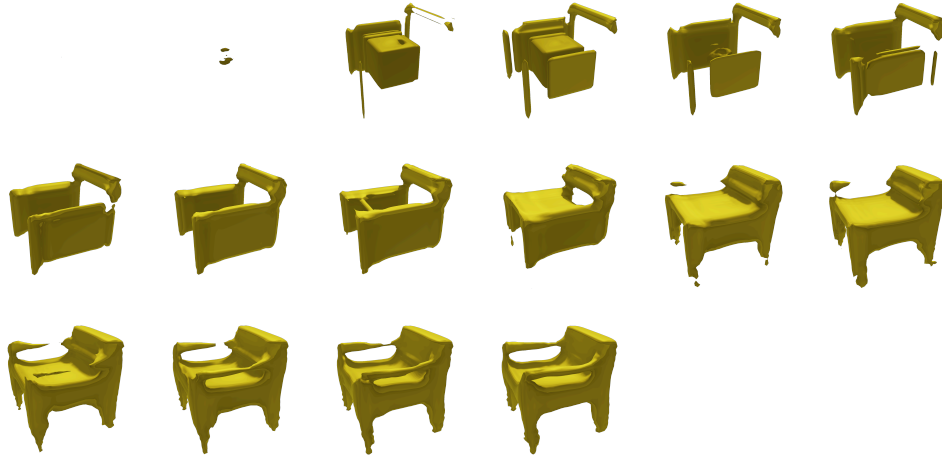


Figure 1.1 Emerging of an SDF chair model sampled from a learned distribution p_{data}

CHAPTER 2

BACKGROUND

This chapter outlines the foundational prerequisites. While formal proofs are omitted to maintain focus on practical and conceptual understanding, appropriate citations are provided for readers seeking rigorous mathematical treatments.

2.1 Cellular Automata

2.1.1 Definition and Basic Principles

Cellular Automata (CA) models were first introduced as self-replicating systems [1] inspired by biological systems. CA are d -dimensional spatio-temporal, locally interacting computational models that obey a handcrafted update rule.

Classical CA models are discrete, deterministic, and synchronous systems that typically operate on a uniform grid of cells $G \subseteq \mathbb{Z}^d$. Here, S_t represents the global state of the grid at time t , and F represents the deterministic update rule, which depends only on the previous state of the grid as formulated in Equation 2.1.

$$S_{t+1} = F(S_t) \tag{2.1}$$

This formulation can be extended by introducing a history of past states [2,3]. Given a memory capacity T , the update rule F can be redesigned to consider the previous $T + 1$ states, as shown in Equation 2.2.

$$S_{t+1} = F(S_t, S_{t-1}, \dots, S_{t-T}) \tag{2.2}$$

This establishes the behavior of the system from a global perspective. To analyze the local and emergent behavior of a Cellular Automaton, it is necessary to examine the underlying dynamics from the perspective of a single cell.

Let $x_t \in S_t$ be the state of a specific cell in the grid $G \subseteq \mathbb{Z}^d$, and let $N_{x_t} = [x_1, x_2, \dots, x_n]$ be the sequence of states of its neighboring cells, where n is the size of the defined neighborhood. The updated state of this individual cell is given in Equation 2.3.

$$x_{t+1} = f(x_t, N_{x_t}) \quad (2.3)$$

Therefore, the global grid state update rule F can be defined as the collective application of the local rule f to all cells, as expressed in Equation 2.4.

$$F(S_t) = \{f(x_t, N_{x_t}) \mid \forall x_t \in S_t\} \quad (2.4)$$

Furthermore, this framework is not limited to scalar representations [4]. By introducing multiple channels the spatial grid expands into a multi-dimensional feature space. Specifically, while the spatial domain remains a subset of $\mathbb{Z}^d$, the global state of the grid becomes a vector field mapping $S_t : \mathbb{Z}^d \rightarrow \mathbb{R}^C$.

The cell state is naturally extended to encompass these multiple channels, defined as a vector $\mathbf{x}_t \in \mathbb{R}^C$. Consequently, the neighborhood of a cell is no longer a sequence of scalars, but a sequence of C -dimensional vectors. Let $\mathbf{N}_{\mathbf{x}_t} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]$ represent this multi-channel neighborhood. The local update rule f is thus redefined to map the current vector state and its vector neighborhood to a new vector state, as shown in Equation 2.5.

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{N}_{\mathbf{x}_t}) \quad (2.5)$$

Similarly, let $\mathbf{S}_t$ represent the global state of the grid where every cell contains a C -dimensional vector. The global update rule F is redefined as the set of all local multi-channel updates applied across the entire grid, expressed in Equation 2.6.

$$\mathbf{S}_{t+1} = F(\mathbf{S}_t) = \{f(\mathbf{x}_t, \mathbf{N}_{\mathbf{x}_t}) \mid \forall \mathbf{x}_t \in \mathbf{S}_t\} \quad (2.6)$$

For a deeper understanding of CA's historical advancements from a perspective of reversibility, conservation laws, decidability questions, computational universality and limit behavior see [5].

2.1.2 From Simple Rules to Complex Systems: Intuition and Examples

Wolfram initiated the first systematic computational and theoretical study of one-dimensional cellular automata [6] and argued the local and global properties of them.

Chosen $d = 1$ for grid $G \subseteq \mathbb{Z}^d$ is a one-dimensional cellular automata, $f(x_t, N_{x_t})$ is shown as a rule table in Table 2.1.

							
							
111	110	101	100	011	010	001	000
0	0	0	0	0	0	0	0

Table 2.1 Rule 0 configuration.

For state **101** $x_t = 0$, $N_{x_t} = [1, 1]$, $n = 2$

For state **011** $x_t = 1$, $N_{x_t} = [0, 1]$, $n = 2$

For state **010** $x_t = 1$, $N_{x_t} = [0, 0]$, $n = 2$

...

And since all the next states $x_{t+1} = 0$ update rule for Table 2.1 which means all the values at the last row are 0, update rule $f(x_t, N_{x_t}) = 0$. And this is why it is called zero rule. Zero rule generates nothing for all possible combinations of initial states as expected.

But let's look at another rule slightly more interesting to look at it create patterns. Let's define update rule f as the following Equation 2.7.

$$\begin{aligned}
 x_t &= x_0, N_{x_t} = [x_1, x_2] \\
 f(x_0, [x_1, x_2]) &= x_0(x_1 + x_2 - 2x_1x_2)
 \end{aligned}
 \tag{2.7}$$

This definition of f generates the Rule 18 in Table 2.2

							
							
111	110	101	100	011	010	001	000
0	0	0	1	0	0	1	0

Table 2.2 Rule 18 configuration.

It gets this name because the last row of Table 2.2 represents number 18 in binary. And this simple rule is the generator of the following patterns under different initial states shown in Figure 2.1.

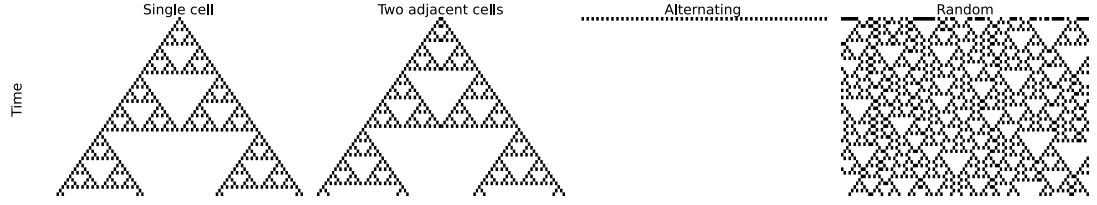


Figure 2.1 Rule 18 with different initial conditions.

We can redefine f to achieve even more chaotic structures. Following definition of f in Equation 2.8 achieves the famous Rule 30

$$f(x_0, [x_1, x_2]) = x_1 + x_0(x_2 - 2x_1x_2) \quad (2.8)$$

which yields the results in Figure 2.2 with the corresponding rule table Table 2.3

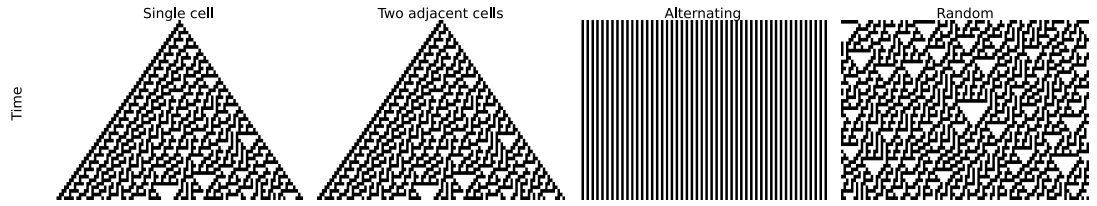


Figure 2.2 Rule 30 with different initial conditions.

Even with the initial condition of starting with a single cell, it is capable of creating patterns where the only way to predict a certain state of a cell x_t in timestep t is to run the system at least t times. Wolfram termed this property *computational irreducibility* [7]. This tells us Rule 30 produces patterns for which no shortcut is known rather than running the simulation.

111	110	101	100	011	010	001	000	
0	0	0	1	1	1	1	1	0

Table 2.3 Rule 30 configuration.

As observed in the study the number of possible configurations the system can actually produce $2^{(2^3)} = 2^8 = 256$ for one-dimensional binary CA. However, a particular CA rule does not necessarily produce all of these configurations during its evolution. The complexity of CA can therefore be studied by examining the set of configurations that can occur and how this set grows with the system size. Wolfram [8] investigated this perspective in detail, introducing entropy and dimension-based measures for characterizing the complexity of CA.

The same idea can be extended to higher dimensional spaces. There exists Conway's Game of Life [9] as one of the most popular two-dimensional example.

Let $G \subseteq \mathbb{Z}^d$, $d = 2$ and $N_{x_t} = [x_1, x_2, \dots, x_n]$, $n = 8$

Game of Life is a two-dimensional binary Cellular Automaton that uses Moore neighborhood (i.e., the eight cells surrounding a cell, including diagonals).

We can define the Game of Life rule table in the following Figure 2.3. Since the rule depends only on the number of alive neighbours showing all other reflections and configurations would be trivial.

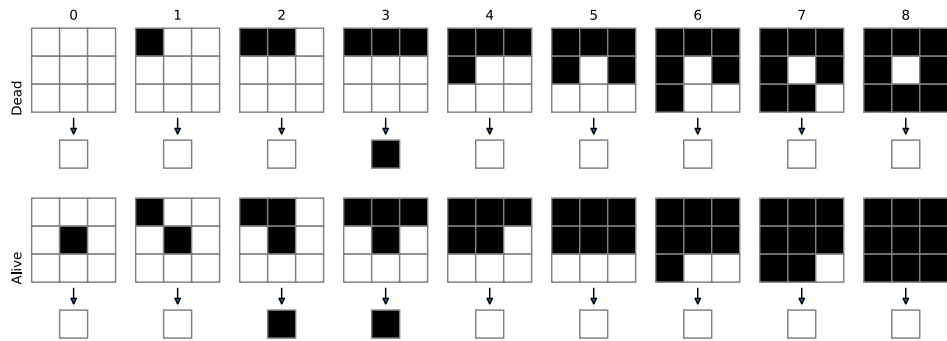


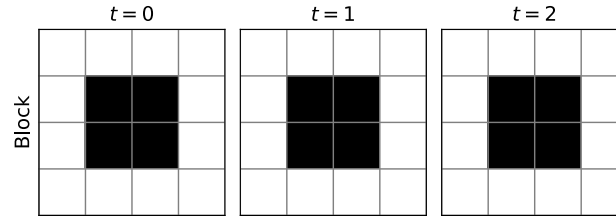
Figure 2.3 Game of Life Rule Table

But it is a more compact and also more widespread representation to define the rule function f as the following partial function shown in Equation 2.9.

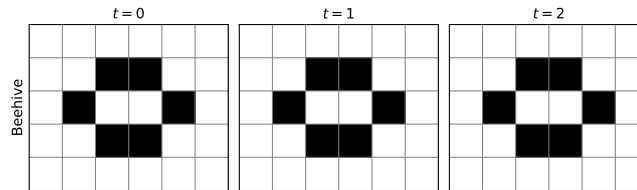
$$n_{\text{alive}} = \sum_{i=1}^8 x_i \quad (\text{number of alive neighbors})$$

$$f(x_0, [x_1, x_2, \dots, x_8]) = \begin{cases} 1 & x_0 = 1 \wedge (n_{\text{alive}} = 2 \vee n_{\text{alive}} = 3) \\ 1 & x_0 = 0 \wedge n_{\text{alive}} = 3 \\ 0 & \text{otherwise} \end{cases} \quad (2.9)$$

Unlike the elementary one-dimensional CA, where all $2^{(2^3)} = 256$ rules can be analysed, the space of two-dimensional rules is much larger ($2^{(2^9)}$ for the Moore neighborhood), and Game of Life is only one particular rule in it. What makes it special is that Conway's Game of Life is proven to be computationally universal [10, 11]. To put it simply, a Turing machine can be simulated using the local construct units introduced by the Game of Life. The units are categorized in different ways based on their behavior.



(a) Block Unit, the most basic static unit.



(b) Beehive Unit, also static.

Figure 2.4 Unless there is external interference, these units do not change.

The units shown in Figure 2.4 are able to remain static by feeding their neighbors by staying alive. When we examine the block in Figure 2.4a we see that all individual cell has exactly 3 neighbors which keep it alive according to the first case in update rule Equation 2.9, and Figure 2.4b also satisfy the same case by having $n_{\text{alive}} = 2 \vee n_{\text{alive}} = 3$.

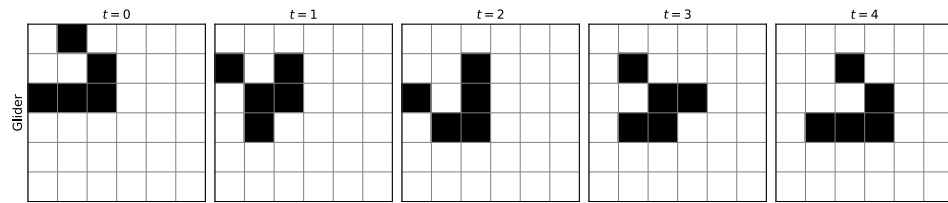


Figure 2.5 Glider Unit, a moving unit that generally represents a bit.

2.2 Neural Cellular Automata

2.2.1 Definition and Basic Principles

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APPENDICES

Appendix A: First Appendix.

Appendix B: Another Table.

Appendix C: Thermophysical Properties of Air.

Appendix D: K_c and K_e Coefficients.

Appendix A – First Appendix

This is a text inside the section

Table A.1 Örnek tablo

A	1
B	2
C	3

Appendix B – Another Table

This is a text inside the section

Table B.1 Örnek tablo

A	1
B	2
C	3

CURRICULUM VITAE

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- | | |
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| - Thermodynamics | - Fluid Mechanics |
| - Heat Transfer | - Internal Combustion Engines |

PRESENT ORGANIZED SOURCES / PRESENTATION

bla bla

**September, 2023
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